

LOCALIZATION IN THE COMMUTANT, STAGE 13:

the leaderboard system read through convex geometry.

The Schur hub as Eldan’s cumulant drift, the critical gain as a conserved dilation charge,
and the network as a difference of support functions

10 October 2026

Abstract

We do three things. First, we take apart the current leaderboard system (v56) and separate what it inherits, what we engineered, and which pieces realize theory. Second, we read six works in full: Klartag–Lehec on the slicing problem, Bizeul–Klartag–Lehec on KLS, Klartag–Ordentlich on the strong data-processing inequality under heat flow, Klartag’s stochastically evolving ellipsoid, Sahasrabudhe’s survey of exponentially small scales, and the caustic-collar paper on billiard rigidity. Third, we prove or check the bridges they open into our setting.

- *The Schur hub is Eldan’s cumulant drift.* Along Eldan’s localization, the drift of the m -th cumulant is $-(m\kappa_m + L_m)$, where L_m contracts pairs of lower cumulants through the inverse covariance (Bizeul–Klartag–Lehec). We show this is the law of total cumulance taken infinitesimally along the localization path. For $m = 4$, L_4 is three $\kappa_3 A^{-1} \kappa_3$ pairings; checked to 10^{-7} on a discrete measure, where it is 20% of the drift. This is the algebraic form of v56’s largest single gain, the Schur-hub path class $12[YC^{-1}Y^\top]_{ii}$.
 - It explains three measured facts: why the hub needs C^{-1} , why it needs the source-resolved (quenched) third cumulant rather than the annealed one, and why its derived weight is right.
 - It predicts the missing pairings for the $(2, 2)$ and $(3, 1)$ slices, with their coefficients.
 - It says κ_3 has no such term ($L_3 = 0$), so the chain’s third-cumulant errors are transport errors, as measured.

In the two-tier language of stage 12, the hub is the between-tier fourth cumulant: v56’s biggest gain is the two-tier framework realized at fourth order.

- *The critical gain is a conserved dilation charge.* RELU commutes with scale mixing, so the network’s common random scale is transported exactly. That is why the mean-direction mode is critical (transport factor ≈ 0.95 measured).
 - Stage 12’s increment law gives its size: across-input log-norm increments $\frac{4}{n}(\frac{3}{2} - \frac{J_2(\theta)}{2\pi}) \approx 4\theta^2/n$, which saturate. This answers the open item of note XLIV with a parameter-free leading law: 0.0058–0.0072 at layers 8–14, against measured 0.0077–0.0100; the input radius gives $1/(2n)$.
 - The gain need not mean-revert to explain the plateau. A falsifiable prediction separates the two readings.
 - The Edgeworth-corrected Gaussian closure is exact to $O(\varepsilon^2)$ on a scale mixture when it carries the mixture’s own skewness $6\varepsilon r$ and kurtosis 12ε (checked: halving ε cuts the error by 3.8–4.0).

So the critical mode leaks into the means only through inconsistency between the chain’s carried variance, κ_3 and κ_4 along the dilation. A dilation-consistent (scale-mixture) closure removes the leak at $O(n^2)$ per layer.

- *The spherical geometry is a support-function geometry.* By homogeneity the problem lives on the sphere.
 - A bias-free RELU network is a difference of support functions of two polytopes (its tropical Newton polytopes), and its Gaussian mean is a difference of mean widths, i.e. first intrinsic volumes. Stage 11’s border formula is the edge formula for this intrinsic volume.
 - Each layer evaluates the support function of the zonoid of the previous layer’s law at the next rows. The Gaussian closure replaces that zonoid by a Gaussian (ellipsoidal) one.
 - The curvature of the zonoid is the wall density of stages 11–12. All three identities are checked exactly.

Klartag’s evolving ellipsoid and the Lovett–Meka walk share one principle: once the process touches a constraint, it keeps it and continues in the constraint-preserving subspace. That principle is the right treatment of the chain’s exact invariants (the dilation charge, the Euler–Stein ladder).

- *What the billiard paper transfers* is the support-function machinery, the grazing parameter $\lambda = t^2$ (our temperature $T \approx \theta^2/2$), and the method of pinning high modes by family invariants. It brings no new algorithm by itself.
- *Heat flow and slicing.* We record a computable memory budget, the negentropy as an integrated localization deficit, checked to 0.5%.

Each claim is labelled as proved, known, measured, conjectural or an analogy. The checks are in `scripts/verify_stage13.py`.

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1 The picture in one page

1. What v56 is (Section 2).

- **The base.** A public chain (504aldo’s v29): Gaussian moment closure, κ_3 sources as CP legs in young and old tiers, hub slices, memoryless κ_4 , a Wick table and Strassen-for-billing. We added cost engineering.
- **Five theory-derived terms on top.** The quenched κ_4 pair class, the scale-mixture “pair”, the fold (exact first-order $(2, 1)$ feedback as an arm displacement), 103 renormalization counterterms, and the chaos-graded κ_4 diagonal (the Schur hub plus the third-chaos star).
- **Where the gain came from.** The two large wins (-10% each, held out) are exact derived terms whose refitted amplitudes stay at 0.95 and 1.007. Shapes chosen by convention never paid.
- **What is not yet inside.** Stages 10–12’s quantitative structure is not inside, and nothing yet treats the critical dilation mode.

2. The Schur hub is Eldan’s cumulant drift (Section 4). Along Eldan’s normalized localization, $d\kappa_m = \kappa_{m+1}(\cdot, A^{-1/2}dB) - (m\kappa_m + L_m)dt$, and we show L_m is the between-path part of the law of total cumulance.

- $L_3 = 0$, so κ_3 ’s errors are transport errors.
- L_4 is three $\kappa_3 A^{-1} \kappa_3$ pairings, three times the variance of the regression of the square on the field. This is the form of v56’s path class $12[YC^{-1}Y^\top]_{ii}$, and it explains why the hub needs C^{-1} , the quenched (source-resolved) κ_3 , and weight 1.
- It predicts the $(2, 2)$ and $(3, 1)$ pairings, with coefficients, from the same solve.
- In the two-tier language, v56’s largest gain is the between-tier fourth cumulant.

3. The critical gain is a conserved dilation charge (Section 5). RELU commutes with scale mixing, so the common scale is transported with factor exactly 1.

- Its size follows stage 12’s increment law: $\frac{4}{n}(\frac{3}{2} - \frac{J_2(\theta)}{2\pi}) \approx 4\theta^2/n$ per layer, saturating at $nV \rightarrow 44.6$; the input radius contributes $1/(2n)$.
- Predicted values are 0.0058–0.0072 at layers 8–14, against measured 0.0070–0.0100.
- The plateau needs no mean reversion; the test is stated.
- The Edgeworth-corrected closure is exact to $O(\varepsilon^2)$ on a scale mixture carrying its own $\gamma_1 = 6\varepsilon r$ and $\gamma_2 = 12\varepsilon$. So the charge moves the means only through inconsistency of the carried variance, κ_3 and κ_4 .
- Design rule: impose the scale-mixture package, at $O(n^2)$ per layer. The truth is about 70% package, with kurtosis about $1.7\times$ the skewness slope against the package’s 2.

4. The spherical geometry (Section 6).

- The network is $h_{P_+} - h_{P_-}$, a difference of support functions of tropical polytopes, and $\mathbb{E}f = (V_1(P_+) - V_1(P_-))/\sqrt{2\pi}$, a difference of first intrinsic volumes. Stage 11’s border formula is the edge formula of V_1 .
- Each layer evaluates the support function of its law’s zonoid at the next rows. The closure is the Gaussian (ellipsoidal) zonoid, and the zonoid’s curvature is the wall density.
- Klartag’s evolving ellipsoid gives a dictionary (contacts as frozen gates, free directions as unresolved gates) and one principle: keep every exact constraint you touch.

5. Billiards, packing, heat flow (Sections 7–8).

- The caustic paper’s support-function envelopes, its positivity $g + g'' > 0$ and its $\lambda = t^2$ are ours: the wall density, and the temperature $T \approx \theta^2/2$.
- Its rigidity has a probabilistic twin: a closure exact on a full family of tilts forces Gaussianity.
- Its Newton-on-high-modes method needs family invariants we lack. That is the flex theorem from the other side; v56 supplied the missing data through source-resolved legs.
- From slicing: the negentropy is an integrated localization deficit, checked to 0.5%, a candidate memory budget.
- From Shannon–Stam stability: the quenched content is the variance of the localized covariance, whose diffusion coefficient is κ_3 .

6. What to build (Section 9).

1. The dilation-consistent closure.
2. The remaining L_4 pairings from v56’s existing solve.
3. Sticky projection onto the exact identities.
4. The memory-budget diagnostic.

2 What v56 is: inherited, engineered, and theory-realized parts

The system is the baked file `estimator_final_v56.py` in `notes/leaderboard-system` on branch `claude/bold-albattani-xgecxi`, made from `est_v29.py` with one configuration string. Scored on all 100 official networks it has adjusted MSE 3.137×10^{-9} in-sample for its counterterms, and 3.166×10^{-9} by the held-out protocol (raw 1.56×10^{-8} , $C/B = 0.203$). That is -24% against our V35 and roughly -42% against the public v29 it started from. The public leaders are quoted at $1.1\text{--}2.1 \times 10^{-9}$ (note XXXIX).

2.1 What it computes, layer by layer

The state carried from layer to layer is a truncated cumulant description of the pre-activations z_l :

- the mean μ_l and the full covariance C_l ;
- third-cumulant *sources*: each layer gives birth to a κ_3 “star” $\text{Sym}(C_b \otimes I \otimes C_b \text{diag}(w_2))$, the Price/Hermite term of ReLU at the Gaussian point, carried as CP-factor legs (A_s, P_s) ;
- a few slices of the fourth cumulant (a diagonal and some $(2, 2)$ and $(3, 1)$ shapes).

One layer then runs these steps:

1. **Linear step.** $\mu \leftarrow W\mu$ and $C \leftarrow WCW^\top$ exactly. The legs are transported by the mean-gate propagator $B = W \text{diag}(\Phi)$, which is first-order linear response, the “product gate”. Young sources stay dense. Old sources are compressed to a shared rank-320/192 Lyapunov basis, re-orthogonalized at one randomized range-finder *join* per layer.
2. **Slices.** The (3) and $(2, 1)$ slices of κ_3 (D3, D21) are read from the legs by Hadamard “hub” products.
3. **Wick stage.** The post-activation moments come from a Gaussian-plus-Edgeworth closure in the carried cumulants: a Hermite/ σ table, Mehler order 2, giving K11, K21, K22, K1–K4.
4. **Birth.** A new source is born from the closure. A rank-4 residual of the $(2, 1)$ slice is kept, and κ_4 is fed back into κ_3 .
5. **Fourth-order sector.** κ_4 is regenerated without memory: annealed transport of the $(2, 2)$ content plus an adaptively fitted λ , with corrections.

All large products are written as Strassen recursions so that flopscope bills their analytic count.

2.2 Inherited, engineered, and theory-realized

component	origin	measured worth
<i>Inherited from the public chain (504aldo v29; probably shared with other leaders)</i>		
Gaussian moment closure with Hermite/Wick tables	standard (the organisers' cumulant-propagation reference)	the backbone
κ_3 sources as transported CP legs, young/old tiers, range-finder joins	504aldo v29	the backbone
D3/D21 hub slices; memoryless κ_4 with fitted λ ; birth residual; feedback	504aldo v29	the backbone
Strassen-for-billing	504aldo v29	the cost model
<i>Engineered by us (cost and numerics, theory-light)</i>		
warm-started joins (Lyapunov continuity), Strassen level 6, symmetric/aliased Grams, billing fixes, dead-neuron saturation drop, leaf minima, join schedule, retuned ranks	notes XV, XXIX, V29–V35	–3 to –6% adjusted from warm joins and level 6; C/B from 0.267 (v29) to 0.180 (V35)
<i>Theory realized in code</i>		
quenched κ_4 pair class (V33.K4Q)	retained transport, note XXVIII: quenched over annealed	–2.3%
the “pair” (V33.WK4M=3, V31.K4D=3)	gain-aware / scale-mixture shapes, notes XI, XIX, XXXVIII	–0.9%
the fold (V52.FB_FOLD=1)	exact first-order (2, 1) feedback as a displacement of the arm, $a \rightarrow a + Y_t/2w_2 + X_t/6$, with the arm's double duty	–10.0% held-out adjusted
counterterms (V47.CAL, 103 amplitudes)	renormalization: fitted elements as counterterms of the closure	–3.8%
chaos-graded κ_4 diagonal (V56.P4)	Wiener chaos grading: Schur-hub path class plus third-chaos star at derived weights	–9.8% held-out adjusted (–20.9% raw)

Three remarks.

1. **The two large wins are exact, derived terms whose fitted amplitude stays at its derived value.** The fold refits at 0.95 and the chaos correction at 1.007. In note XXXIX's sense they are law terms. The conventions (shapes of omitted classes chosen from the pair state) never paid, as the flex theorem predicted.

2. **Where the programme's ideas are already inside.**

- The quenched/annealed distinction of stage 9 is realized twice. The quenched κ_4 pair class wins, and in the chaos correction the source-resolved hub product wins while its symmetrized (annealed) control carries nothing.
- The quasi-free (Fock/Bogoliubov) sector of stages 11–12 is realized as the chaos grading.
- Section 4 shows that the Schur hub is the between-tier fourth cumulant of the two-tier (localization) decomposition.

3. What is not inside yet.

- Nothing of stages 10–12’s quantitative structure: the Mehler clock, the parabolic depth, the internal Eldan clock and temperature, the hyperbolic upper tier, the wall-density law.
- In particular, nothing yet treats the critical dilation mode that note XLIV measured. Section 5 supplies its law and a closure that respects it.

3 What the six works do

Klartag–Lehec, the slicing problem.

- **Localization setup.** For an isotropic log-concave μ , localization tilts by $e^{\theta \cdot x - t|x|^2/2}$ with $d\theta_t = dW_t + a_t dt$. The covariance process A_t obeys $\mathbb{E}A_t + \mathbb{E}a_t \otimes a_t = \text{Id}$ (the variance decomposition) and $\frac{d}{dt}\mathbb{E}A_t = -\mathbb{E}A_t^2$.
- **The final step.** Guan’s bound $\mathbb{E} \text{Tr} A_t^2 \leq Cn$ plus a projection-monotonicity lemma produce a subspace and a time $t_0 \sim 1$ with $\mathbb{E}A_{t_0} \geq \frac{1}{4}\text{Id}$, without losing the isotropic constant (Milman’s M -ellipsoids). Eldan–Mikulincer’s stability of Shannon–Stam then bounds $\int \mathbb{E}|\Gamma_r - \mathbb{E}\Gamma_r|^2/(1-r) dr$, the fluctuation of the localized covariance, by the entropy gain under $X \mapsto (X + X')/\sqrt{2}$, which Ball–Nguyen bound by $2n$.
- **The identity we use** (their Corollary 3.2 with Lemma 3.3): the relative entropy to the Gaussian is the integrated Föllmer drift, which in localization time reads

$$D(\mu \parallel \gamma) = \frac{1}{2} \int_0^\infty \left(\frac{n}{1+t} - \mathbb{E} \text{Tr} A_t \right) dt \quad (\mu \text{ isotropic}).$$

Bizeul–Klartag–Lehec, KLS.

- **The base case.** The third-cumulant bound $R_3 \leq 8\text{Id}$ comes from the sharp generalized thin shell.
- **The induction.** It is propagated to $R_m \leq K^{m-1}((m-1)!)^2\text{Id}$ through Eldan’s normalized localization $dc = A^{-1}a dt + A^{-1/2}dB$, $dQ = A^{-1}dt$. Its key *cumulant equation* (their Lemma 4.3) is

$$\begin{aligned} d\kappa_m &= \kappa_{m+1}(\cdot, \dots, \cdot, A_t^{-1/2}dB_t) - (m\kappa_m + L_m)dt, \\ L_m(h) &= \sum_{\substack{J \ni 1 \\ 2 \leq |J| \leq m-2}} \langle \kappa_{|J|+1}(h_J, \cdot), A_t^{-1}\kappa_{m-|J|+1}(h_{J^c}, \cdot) \rangle. \end{aligned}$$

- **The finish.** A suspension turns cumulant bounds into bounds on the Taylor tensors of tilt averages $F_f(z) = \int f e^{\langle z, x \rangle} d\mu / \int e^{\langle z, x \rangle} d\mu$. The Song–Zhang criterion then gives a dimension-free Poincaré constant.
- **The Gaussian case.** For the Gaussian input, $F_f(z) = \mathbb{E}f(X+z)$ is exactly stage 11’s Husimi symbol at $\hbar = 1$, and its Taylor tensors are the Wiener chaos kernels that v56 grades.

Klartag–Ordentlich, SDPI under heat flow.

- **The contraction coefficient.** For $Y = X + \sqrt{s}Z$, the χ^2 coefficient is the squared maximal correlation $S^2(X, Y)$. It is bounded between $1 - \text{mmse}/\mathbb{E}|X - \mathbb{E}X|^2$ and $(1 + s/C_P)^{-1}$, and is at least e^{-s/C_P} for log-concave X .
- **Log-convexity.** For log-concave X , $s \mapsto \log \text{Var}(\mathbb{E}[f(X)|Y])$ is convex.
- **Two-flow de Bruijn.** $\frac{d}{ds}D(\nu_s||\mu_s) = -\frac{1}{2}J(\nu_s||\mu_s)$, with both measures under the flow.

Klartag, the evolving ellipsoid.

- **The process.** Start from a ball free of points of a random lattice. Run a Dyson Brownian motion $dA_t = \pi_t(dW_t)$ on the ellipsoid matrix, where π_t projects onto $\{B : Bx \cdot x = 0 \text{ for every contact point } x\}$. Every lattice point the ellipsoid touches stays on its boundary.
- **The volume accounting.** By Itô,

$$d \log \det A_t = \langle A_t^{-1}, \pi_t dW_t \rangle - \frac{1}{2} \sum_{i,j} \frac{|\pi_t(u_i \otimes_s u_j)|^2}{\lambda_i \lambda_j} dt,$$

and the drift is at most $-\dim F_t/(2\|A_t\|^2)$. Here $\dim F_t \geq n(n+1)/2 - |\text{contacts}|/2$.

- **The count.** Contacts accrue at rate about $e^{n^{2t/8}}$ (Siegel’s formula and the reflection principle). Maurey’s trick $A_t - a_0 = \frac{1}{2}(W^1 + W^2)$ controls $\|A_t\|$. At $T = 16n^{-2} \log n$ this gives $\mathbb{E} \log \det A_T \leq C - 4 \log n$: an empty ellipsoid of volume cn^2 , hence lattice packings of density $cn^2 2^{-n}$.

Sahasrabudhe, exponentially small scales.

- **The common thread.** A random process is guided toward an exponentially unlikely target. Examples: the Lovett–Meka walk, a Brownian motion that sticks to every constraint slab it hits and continues inside it; the Rödl nibble for independent sets; Klartag’s ellipsoid.
- **Codegree control.** Campos–Jenssen–Michelen–Sahasrabudhe get $(1 - o(1)) \frac{d \log d}{2d+1}$ packings and spherical codes from a nibble on a Poisson-discretized graph. Control of *codegrees*, not triangles, makes the random-like heuristic exact.
- **Random symmetric matrices.** Inverse Littlewood–Offord theory and approximate negative correlation give an e^{-cn} singularity bound for random symmetric ± 1 matrices.

The caustic-collar billiard paper. A smooth strictly convex positively curved table whose grazing annulus is continuously foliated by invariant essential curves is an ellipse. The proof runs in four steps.

1. **Graphs and periodic leaves.** Twist and area recurrence make every leaf a graph $p = g_b(\theta)$ in line coordinates, and make every small rational rotation l/q a wholly periodic leaf.
2. **Action constancy.** A closed stationary chain on a graph has its action constant in the starting point. Its $\pm q$ Fourier rows $C_q(u)$ therefore vanish. Their derivative is the identity on high modes plus $O(1/q)$, so a Newton scheme on shrinking complex strips forces the curvature datum u to be analytic.

3. **Joint analyticity.** A jointly analytic conjugacy $V(y, t)$ is built across the rational resonances. The periodic leaves supply the compatibility of residual values and of first parameter jets.
4. **Caustics.** The analytic line graphs are support functions $g(\theta, \lambda)$, with $\lambda = t^2$, $g + g_{\theta\theta} > 0$ and $g_\lambda < 0$. Their envelopes $\Gamma_\lambda = gn_\theta + g_\theta e_\theta$ are nested convex caustics filling a collar, and the companion theorem concludes.

4 The Schur hub is Eldan's cumulant drift

4.1 Eldan's cumulant hierarchy is the law of total cumulance in time

Let μ be a compactly supported probability measure on $\mathbb{R}^d$. Run Eldan's normalized localization: $\mu_t \propto e^{\langle c_t, x \rangle - \frac{1}{2} \langle Q_t x, x \rangle} \mu$ with $dc = A_t^{-1} a_t dt + A_t^{-1/2} dB_t$ and $dQ = A_t^{-1} dt$. Write $\kappa_m(t)$ for the cumulant tensors of μ_t .

Proposition 4.1 (Bizeul–Klartag–Lehec, Lemma 4.3; known). $d\kappa_m = \kappa_{m+1}(\cdot, \dots, \cdot, A_t^{-1/2} dB_t) - (m\kappa_m + L_m) dt$, with L_m as in Section 3. In particular:

- $L_3 = 0$, so $\mathbb{E}\kappa_3$ only decays, at rate 3.
- The fourth-order term is

$$L_4(h_1, \dots, h_4) = \sum_{j=2}^4 \langle \kappa_3(h_1, h_j, \cdot), A^{-1} \kappa_3(h_{\{2,3,4\} \setminus \{j\}}, \cdot) \rangle.$$

For instance, $L_4(u, u, u, u) = 3 \langle \kappa_3(u, u, \cdot), A^{-1} \kappa_3(u, u, \cdot) \rangle$.

Proposition 4.2 (the drift is total cumulance; proved). μ_{dt} is the posterior of $X \sim \mu$ given one infinitesimal Gaussian observation. Brillinger's law of total cumulance over set partitions π ,

$$\kappa_m(X) = \sum_{\pi} \kappa_{|\pi|}(\kappa_B(X | \mathcal{G}))_{B \in \pi},$$

applied to that observation, reproduces the drift $-(m\kappa_m + L_m)$ exactly:

- the one-block partition gives $\mathbb{E}\kappa_m(dt)$;
- two-block partitions with a singleton give $m\kappa_m dt$;
- two-block partitions with both blocks of size ≥ 2 give $L_m dt$;
- partitions into three or more blocks contribute $o(dt)$.

Proof. Conditionally on the observation, $\kappa_k(dt) = \kappa_k + \kappa_{k+1}(\cdot, A^{-1/2} dB) + O(dt)$. The covariance over dB of the fluctuations of $\kappa_{|B_1|}$ and $\kappa_{|B_2|}$ is $\langle \kappa_{|B_1|+1}, A^{-1} \kappa_{|B_2|+1} \rangle dt$. A singleton block $B_1 = \{i\}$ has $\kappa_2 = A$, and $\langle Ah_i, A^{-1} \kappa_m(h_{\text{rest}}, \cdot) \rangle = \kappa_m(h)$. Joint cumulants of three or more $O(\sqrt{dt})$ Gaussian-linear fluctuations vanish at order dt . $\square$

So L_m is the *between-path* part of the m -th cumulant: what conditioning on the localization path moves out of the within-path cumulants. It is quadratic in lower cumulants, and the metric is the inverse covariance, because the observation noise is $A^{-1/2} dB$. The identity is checked on a 7-point measure in $\mathbb{R}^3$ by Gauss–Hermite averaging over one step (Richardson in dt).

- The drift of κ_3 matches $-3\kappa_3$ to 1.5×10^{-7} .
- The drift of κ_4 matches $-(4\kappa_4 + L_4)$ to 2.8×10^{-7} ; dropping L_4 errs by 21%.
- $L_4(u^4)$ equals $3\langle \kappa_3(u, u, \cdot), A^{-1}\kappa_3(u, u, \cdot) \rangle$ to all digits.

4.2 The Schur hub is the between-tier fourth cumulant

Proposition 4.3 (static form; proved). *For any σ -algebra $\mathcal{G}$ and scalar X ,*

$$\begin{aligned} \kappa_4(X) = \mathbb{E}\kappa_4(X|\mathcal{G}) + 4 \text{Cov}(\mathbb{E}[X|\mathcal{G}], \kappa_3(X|\mathcal{G})) + 3 \text{Var}(\text{Var}(X|\mathcal{G})) \\ + 6 \kappa(\mathbb{E}[X|\mathcal{G}], \mathbb{E}[X|\mathcal{G}], \text{Var}(X|\mathcal{G})) + \kappa_4(\mathbb{E}[X|\mathcal{G}]). \end{aligned}$$

If $\mathcal{G} \supseteq \sigma(Y)$ for a vector Y with covariance C_Y , then the projection inequality gives

$$3 \text{Var}(\text{Var}(X|\mathcal{G})) \geq 3 \text{Cov}(\text{Var}(X|\mathcal{G}), Y) C_Y^{-1} \text{Cov}(Y, \text{Var}(X|\mathcal{G})).$$

Equality holds when the conditional variance is affine in Y .

Read in the two-tier language of stage 12:

- the upper tier is the conditioning ($\mathcal{G}$, the localization centre, i.e. the field the next layer sees);
- the lower tier is the conditional law;
- the term $3 \text{Var}(\text{Var}(X|\mathcal{G}))$ is the *between-tier* fourth cumulant, and its Schur projection is the $\kappa_3 A^{-1} \kappa_3$ form of L_4 .

The hierarchy's term has the same shape. For centred X , $\kappa_3(u, u, \cdot) = \text{Cov}((u \cdot X)^2, X)$, so

$$L_4(u^{\otimes 4}) = 3 \text{Cov}((u \cdot X)^2, X) A^{-1} \text{Cov}(X, (u \cdot X)^2) = 3 \text{Var}(\text{proj}_{\text{span} X}(u \cdot X)^2)$$

under μ_t , averaged over paths: three times the variance of the linear regression of the square on the field. v56's path class is a regression of the same kind. Note XLI writes it as

$$P4_i = 12 [Y C^{-1} Y^\top]_{ii} = 3 \text{Var}(\text{proj}_{\text{span } z_l} C_i), \quad 2Y_{ij} = \text{Cov}(C_i, z_{l,j}),$$

where C_i is the cross term between neuron i 's first and second chaos. Using the cross term instead of the whole square removes two within-tier pieces: the square of the first chaos, which the Gaussian closure already handles, and the square of the second chaos, the 4-cycle.

In the chaos-2 model $z = L \cdot x + \frac{1}{2}(x^\top H x - \text{tr} H)$, the cross term projects onto the input's linear span as $2(HL) \cdot x$, so 3Var of it is $12|HL|^2$, the path class of $\kappa_4 = 12|HL|^2 + 3 \text{tr} H^4$. Under localization this projection is the part of the conditional variance that moves linearly with the centre. Indeed $\text{Var}(z|\mathcal{F}_t) = |L + Hm_t|^2/(1+t) + \text{tr} H^2/2(1+t)^2$, whose linear fluctuation is $2L^\top Hm_t/(1+t)$. So the path class is the between-tier part of the fourth cumulant, read through the field.

Corollary 4.4 (what it explains in v56; interpretation, with note XLI's data). *1. **The metric is C^{-1} , **not** $\text{diag}(C)^{-1}$. The observation noise of localization is $A^{-1/2}dB$. Measured: the hub with the diagonal metric is 30–60 times too large.***

2. **The cumulants must be conditional (quenched).** The drift involves $\mathbb{E}[\kappa_3(t)A_t^{-1}\kappa_3(t)]$ over paths, not $\mathbb{E}\kappa_3 A^{-1}\mathbb{E}\kappa_3$. The annealed slice $D21 = \kappa(z_i, z_i, \cdot)$ mixes within-tier and between-tier pieces. Measured: the symmetrized-D21 control removes 0–6%, the source-resolved hub 64–81%.
3. **The weight is fixed.** The three pairings of $L_4(e_i^{\otimes 4})$ are the 3 in 3 Var. Measured: free-running prefers the derived weight 1, and renormalization leaves it at 1.007.
4. **Each half alone is adverse.** Proposition 4.3 has five terms, and the third-chaos star is the $4\text{Cov}(\mathbb{E}, \kappa_3)$ -type partner. Measured: either piece alone is +79% to +131%; together they give –21%.

4.3 Predictions: the other pairings from the same solve

The hierarchy gives every slice of L_4 , with coefficients:

$$\begin{aligned} L_4(e_i, e_i, e_j, e_j) &= \langle D_i, A^{-1}D_j \rangle + 2\langle \kappa_3(e_i, e_j, \cdot), A^{-1}\kappa_3(e_i, e_j, \cdot) \rangle, \\ L_4(e_i, e_i, e_i, e_j) &= 3\langle D_i, A^{-1}\kappa_3(e_i, e_j, \cdot) \rangle, \end{aligned}$$

with $D_i = \kappa_3(e_i, e_i, \cdot)$. In v56’s dressed form the first term of the (2,2) slice is note XLI’s “cheap path part” $4[YM^{-1}Y^\top]_{ij}$ (one product from the existing solve). The second term and the (3,1) slice need the off-diagonal third cumulant contracted once: $\sum_k \kappa_3(i, j, k) y_{ik}$ with $y_i = C^{-1}D_i$ already formed. On the carried CP sources this is one hub-type contraction per source-layer, about the cost of the existing D21 hub.

Conjecture 4.5 (testable on note XLI’s dumps). *The source-resolved (3,1) pairing*

$$3 \sum_k \kappa_3(i, j, k) [C^{-1}Y]_{ik}$$

removes a substantial part of the chain’s K31 error. That error is 74% rms; about 40% of the (3,1) slice is “history”, not a function of the pair state. The pairing uses the full source tensor, not the pair state, so it is exactly the kind of term the flex theorem allows to carry new information.

Remark 4.6 (κ_3 has no Schur term). $L_3 = 0$: under localization the expected third cumulant only decays. A per-layer closure of κ_3 therefore has no between-tier correction to add. Its errors are transport errors. This matches note XXXIX: the chain’s D21 error is the flat incoherent sector of the transport (the product-gate defect), right to 2% in its coherent part.

5 The critical gain is a conserved dilation charge

Note XLIV (section 7) located the largest structured part of v56’s late covariance error. It is one number per layer: the eigenvalue of the pre-activation covariance along the mean direction, biased low by 0.5–0.8%. This mode is the dilation (gain), its error is transported with factor about 0.95, and its relative variance τ is roughly stationary at 0.008–0.010 from layer 6. Note XLIV left one theory item open: an exact law for how the gain’s increments compose through a RELU layer. Here it is.

5.1 The charge

Proposition 5.1 (the dilation charge is conserved; proved). *Let $z = (1 + \delta)g$ with δ independent of g , $\mathbb{E}\delta = 0$, $\text{Var } \delta = \varepsilon$ and $1 + \delta > 0$. Then $\text{relu}(z) = (1 + \delta)\text{relu}(g)$, and the next layer is $W\text{relu}(z) = (1 + \delta)W\text{relu}(g)$. The family of common-scale mixtures is invariant under the layer map, and the mixing variance ε is transported unchanged. In covariance form, $\text{Cov}(z) = (1 + \varepsilon)\text{Cov}(g) + \varepsilon\mathbb{E}g\mathbb{E}g^\top$: the charge lives on the mean direction.*

This is the exact reason the mode is critical. An error in ε is carried with factor exactly 1, while every other direction is contracted: a layer-3 injection loses about $9\times$ in squared norm by layer 15 (note XLII). Two other views say the same.

- In stage 12’s hyperbolic upper tier, dilation is translation along the kink geodesic, and He initialization is the unique scale with zero drift along it.
- In the information-theoretic language of Klartag–Ordentlich (an analogy): a channel contracts every perturbation it does not preserve, and the layer map preserves the scale-mixture family exactly. The dilation orbit is the one direction with contraction coefficient 1.

5.2 The size law

Proposition 5.2 (the dilation sector’s leading law; stage 12, Proposition 3.3, at leading order in $1/n$). *Let $V_l = \text{Var}_x \log |h_l|^2$ for a fixed network. Its across-input increments are*

$$V_l - V_{l-1} = \frac{4}{n} \left(\frac{3}{2} - \frac{J_2(\theta_{l-1})}{2\pi} \right) = \frac{4\theta_{l-1}^2}{n} + O(\theta^4/n), \quad V_0 = \frac{2}{n},$$

so $nV_l \rightarrow 44.6$. *The relative variance of the mean-direction mode is*

$$\tau_l \simeq \frac{V_l}{4} + \frac{1}{nt_l},$$

where the second term is the bulk’s own projection on the mean direction and t_l is the internal clock.

At $n = 1024$ the leading law gives $V_l/4 = 0.0051, 0.0058, 0.0064, 0.0068, 0.0072$ at $l = 6, 8, 10, 12, 14$, with limit 0.0109. The input radius contributes $2/(4n) = 4.9 \times 10^{-4}$, which is note XLIV’s “ $1/(2n)$ ”. Measured values are:

- on stage 12’s network (seed 5), the Rayleigh quotient along the mean is 0.0070, 0.0077, 0.0085 at $l = 8, 12, 16$, with measured V_l about 7% above the leading law;
- note XLIV’s τ , which uses the top eigenvalue and so sits above the Rayleigh quotient, is 0.0087–0.0100 (network 0) and 0.0077–0.0094 (network 1) at layers 6–14.

So the law has the right order, sign and slow saturation, and is 20–30% low at leading order. The quenched value for a given network is an $O(n^2)$ sum: $\sum_{a,b} \text{Cov}(\text{relu}(z_a)^2, \text{relu}(z_b)^2)$ over the chain’s own correlations, a bivariate degree-two arc-cosine kernel per pair.

Remark 5.3 (saturation, not mean reversion; a falsifiable prediction). Note XLIV read the stationary τ as a mean-reverting gain. The increment law gives a different mechanism. The gain’s across-input increments are independent to leading order: the weights are fresh at every layer. Their variance $4\theta^2/n$ is summable, because two inputs at angle θ disagree on only a fraction θ/π of the gates, so deep layers act alike on all inputs. τ therefore saturates without any restoring force.

- *Test*: the across-input correlation of successive log-norm increments $\Delta_l = \log(|h_l|^2/|h_{l-1}|^2)$.
- *Prediction*: near 0, with $\text{Var } \Delta_l \approx 4\theta_{l-1}^2/n$.
- Mean reversion would make the correlation negative.

5.3 When the charge moves the means

Theorem 5.4 (first-order cancellation; proved, and checked). *Let $z = (1 + \delta)(\mu + s\xi)$ with $\xi \sim N(0, 1)$, δ independent and symmetric, $\text{Var } \delta = \varepsilon$ and $1 + \delta > 0$, and put $r = \mu/s$. The Edgeworth-corrected Gaussian closure uses the mixture's own variance $S^2 = s^2(1 + \varepsilon) + \varepsilon\mu^2$, skewness γ_1 and excess kurtosis γ_2 :*

$$S \left[m_1(R) - \frac{\gamma_1}{6} \text{He}_1(R) \varphi(R) + \frac{\gamma_2}{24} \text{He}_2(R) \varphi(R) \right], \quad R = \mu/S.$$

It equals the exact $\mathbb{E} \text{relu } z = s m_1(r)$ up to $O(\varepsilon^2)$. To first order the three corrections are

$$\underbrace{\frac{\varepsilon}{2}(1+r^2)}_{\text{variance}} \underbrace{-\varepsilon r^2}_{\gamma_1=6\varepsilon r} + \underbrace{\frac{\varepsilon}{2}(r^2-1)}_{\gamma_2=12\varepsilon} = 0 \quad (\text{in units of } s\varphi(r)).$$

Proof. $\kappa_3 = 6\mu s^2\varepsilon$ and $\kappa_4 = 12s^4\varepsilon + O(\varepsilon^2)$, by expanding $z - \mathbb{E}z = \delta\mu + s\xi + \delta s\xi$. Then $\partial_{S^2}[Sm_1(\mu/S)] = \varphi/2S$, $\gamma_1 = 6\varepsilon r + O(\varepsilon^2)$ and $\gamma_2 = 12\varepsilon + O(\varepsilon^2)$. Collect the $O(\varepsilon)$ terms. $\square$

Checked for two-point and uniform scale laws at three (μ, s) : the closure error falls by 3.8–4.0 each time ε halves ($1.1 \times 10^{-4} \rightarrow 2.8 \times 10^{-5} \rightarrow 7.0 \times 10^{-6}$ at $\mu = 0.7, s = 1$).

Corollary 5.5 (the design rule; derived). *The critical mode moves the neuron means only through inconsistency. The dilation charge must enter the variance, the third cumulant and the fourth cumulant as one scale-mixture package. To first order in ε and in the chain's notation, with $\tilde{C}$ the non-dilation covariance:*

$$\Delta C = \varepsilon(\tilde{C} + \mu\mu^\top), \quad \Delta\kappa_3(a, b, c) = 2\varepsilon(\mu_a\tilde{C}_{bc} + \mu_b\tilde{C}_{ac} + \mu_c\tilde{C}_{ab}), \quad \Delta\kappa_4(a, a, a, a) = 12\varepsilon\tilde{C}_{aa}^2.$$

In particular $\Delta D3_a = 6\varepsilon\mu_a\tilde{C}_{aa}$ and $\Delta D21_{ac} = 2\varepsilon(2\mu_a\tilde{C}_{ac} + \mu_c\tilde{C}_{aa})$. v56 computes these three pieces by three different routes:

- the covariance exactly, as $WCW^\top$;
- κ_3 through transported legs and births;
- κ_4 through the memoryless regeneration.

Their dilation parts are therefore mutually inconsistent at the measured 0.5–0.8%. Note XLIV's injection ledger says the same: D21 under-feeds the mode at every layer and K31 partly compensates. The rule is to carry ε_l explicitly and write the three package terms from $(\varepsilon_l, \mu_l, \tilde{C}_l)$ in place of the chain's own dilation components. The cost is $O(n^2)$ per layer.

Remark 5.6 (the true law is 70% package; measured). On stage 12's network the truth's marginal cumulants follow the package law in form. The skewness is linear in r ($R^2 = 0.86$ – 0.91), and the excess kurtosis is flat. The amplitudes are $\eta = 0.71$ – 0.72 (skewness) and $\eta' = 0.57$ – 0.63 (kurtosis) of the full- $V/4$ prediction. The package predicts excess kurtosis = $2 \times$ the skewness slope. Measured, the ratio is 1.59, 1.75, 1.77 at $l = 8, 12, 16$. So the truth contains one approximately consistent scale-mixture component of size $\varepsilon \approx 0.005$ at $l = 16$, against $V/4 \approx 0.008$. The chain can estimate it without truth: regress its own γ_1 on r , set $\varepsilon = \text{slope}/6$, and impose the package. Deriving η is open.

Remark 5.7 (Klartag’s principle, applied). Klartag’s ellipsoid and the Lovett–Meka walk keep every constraint they touch and continue in the constraint-preserving subspace. The dilation package is such a constraint. Along it the chain’s transport has no damping, so it must be held exactly rather than approximated. Every other direction forgets its errors.

6 The spherical geometry: support functions, zonoids and mean width

6.1 The network is a difference of support functions

A bias-free RELU network is positively homogeneous: $f(x) = |x|f(\hat{x})$. Under γ the radius and the direction are independent, so $\mathbb{E}f = \mathbb{E}|x| \cdot \int_{S^{n-1}} f d\sigma$: the problem lives on the sphere. Every positively homogeneous piecewise-linear function is a difference of support functions of polytopes (the tropical description of Zhang–Naitzat–Lim). Write $h_K(x) = \max_{y \in K} \langle y, x \rangle$.

Proposition 6.1 (tropical decomposition; proved). $\text{relu}(h_A - h_B) = h_{\text{conv}(A \cup B)} - h_B$. Hence, inductively, $f = h_{P_+} - h_{P_-}$ for two polytopes $P_{\pm}$ built from the rows by Minkowski sums and convex hulls. A first hidden layer contributes zonotopes $\sum_a c_a [0, w_a]$.

Theorem 6.2 (the mean is a difference of first intrinsic volumes; proved).

$$\mathbb{E}f = \frac{\mathbb{E}|x|}{2} (w(P_+) - w(P_-)) = \frac{V_1(P_+) - V_1(P_-)}{\sqrt{2\pi}},$$

where w is the mean width and V_1 the first intrinsic volume. Moreover $V_1(P) = \sum_{\text{edges } e} |e| \gamma(e, P)$, the edge length times the external angle. Under the normal-fan duality this is stage 11’s border formula, $\mathbb{E}f = (2\pi)^{-1/2} \sum_F \kappa_F \alpha_F$:

- the cells of the activation tiling are normal cones of vertices;
- the walls are normal cones of edges;
- the kink κ_F is the edge length $|e|$;
- α_F is the external angle.

Proof. $\mathbb{E}h_K(x) = \mathbb{E} \sup_{y \in K} \langle y, x \rangle = V_1(K)/\sqrt{2\pi}$ (Sudakov–Tsirelson), and polar coordinates give the mean-width form. The edge formula is standard (Schneider, Section 4.2). $\square$

Checked in $\mathbb{R}^2$ with 8 hidden units and a RELU output.

- The identity of Proposition 6.1 holds to 6×10^{-16} .
- The quadrature mean 0.16269286 equals $\frac{\mathbb{E}|x|}{2} \cdot \frac{\text{per}(P_+) - \text{per}(P_-)}{\pi}$ to all printed digits.

The packing and convex-geometry literature is therefore natively about our objects. A network is a pair of polytopes, its mean is an intrinsic volume, its walls are a normal fan, and its gates are a hyperplane tessellation of the sphere. For the last, two inputs at angle θ are separated by a fraction θ/π of the hyperplanes (stage 11).

6.2 Each layer evaluates the support function of a zonoid

For a law μ on $\mathbb{R}^n$, the zonoid $Z(\mu) = \{\mathbb{E}[Xg(X)] : 0 \leq g \leq 1\}$ (Koshevoy–Mosler) has support function $h_{Z(\mu)}(w) = \mathbb{E}\langle w, X \rangle_+$.

Proposition 6.3 (the layer map is support-function evaluation; proved). $\mathbb{E}h_{l,a} = h_{Z(\mu_{l-1})}(w_{l,a})$. The next layer’s means are the values of one convex body’s support function at the n rows. The Gaussian closure replaces $Z(\mu_{l-1})$ by the zonoid of $N(\mu, C)$, whose support function is $sm_1(r)$. For centred laws this zonoid is the ellipsoid $C^{1/2}B/\sqrt{2\pi}$. In general it is an “egg” interpolating between that ellipsoid and the segment to the mean.

Proposition 6.4 (the curvature of the zonoid is the wall density; proved). $\nabla_w^2 h_{Z(\mu)}(w) = \mathbb{E}[\delta(\langle w, X \rangle) X \otimes X]$. In the plane, the radius of curvature $h + h''$ is the density of the projected law at the wall, weighted by the transverse second moment.

Checked for a three-component Gaussian mixture in $\mathbb{R}^2$: the finite-difference Hessian and the wall integral agree to six digits. This joins three pictures:

- stage 12’s wall-density law (the lower tier’s activity is the zonoid’s curvature);
- stage 11’s boundary layer;
- the billiard paper’s positivity $g + g_{\theta\theta} > 0$.

In a 2-plane, the angular harmonics of h_Z away from the Gaussian egg are the cumulant corrections by order: κ_3 feeds odd harmonics and κ_4 even ones. Non-Gaussianity is non-ellipticity of zonoids.

6.3 Klartag’s ellipsoid in our coordinates (dictionary; analogy, labelled)

Klartag’s process	the network and the chain
ellipsoid $\{A_t x \cdot x < 1\}$, $n(n+1)/2$ degrees of freedom	the closure’s Gaussian state (μ, C) ; its $n(n+1)/2$ covariance degrees of freedom
lattice point on the boundary (contact)	a frozen (resolved) gate, $ r_a \gg 1$; the ball no longer crosses that wall
free directions F_t , $\dim F_t \geq$ $n(n+1)/2 - \text{contacts} /2$	unresolved gates, a fraction $\theta/\pi \approx 3/\tau$ per layer (stage 12)
drift of $\log \det A_t$, $-\dim F_t/2 \ A\ ^2$	the information rate of the gate filtration: log-volume of the input cell lost per layer
contacts accrue at rate about $e^{n^2 t/8}$	gates freeze like $\sqrt{T}$, with $T \approx \theta^2/2$ the internal temperature
sticky constraint $\langle A_t, x \otimes x \rangle = 1$	an exact invariant the chain must keep: the dilation package, the Euler–Stein ladder

The structural content is the sticky principle. A process that is only approximately right gains most by holding exactly the constraints along which it cannot damp its own errors. In Klartag’s case these are the contacts, which keep the ellipsoid lattice-free. In ours they are the critical (charge) directions of Section 5.

Spherical codes and codegrees (analogy). The rows of W_1 form a random spherical code, and the first layer’s cells are the 2^n images of orthants. In the packing work of Campos, Jenssen, Michelen and Sahasrabudhe, the nibble’s “random-like” heuristic is made exact by controlling *codegrees*. In ours, the closure (independent, dichotomized-Gaussian gates) is the random-like heuristic, and it is accurate when pair couplings are small. Stage 12 found the bulk couplings $O(n^{-1/2})$ and incoherent. The one coherent, uncontrolled “codegree” is the Mattis (dilation) component. It is again the charge.

7 Billiards, packing and exponentially small scales: what transfers

7.1 The billiard paper, read against our objects

1. **The machinery is ours already.** The paper’s last step turns invariant line graphs $p = g(\theta, \lambda)$ into convex bodies through support functions. Three of its objects are ours.

- *Contact points.* Its envelope $\Gamma_\lambda = g n_\theta + g_\theta e_\theta = \nabla h$ is the contact point of each support line. For a zonoid, $\nabla_w h_{Z(\mu)}(w) = \mathbb{E}[X \mathbf{1}\{\langle w, X \rangle > 0\}]$ is the gate-weighted mean, i.e. neuron w ’s first-chaos vector, which is the chain’s “arm”.
- *Convexity.* Its condition $g + g_{\theta\theta} > 0$ is Proposition 6.4: positivity of the wall density.
- *Nesting.* Its condition $g_\lambda < 0$ says the family moves inward. In our setting the zonoids of localized laws are a martingale family: $\mathbb{E}h_{Z(\mu_t)} = h_{Z(\mu)}$, while each path’s zonoid shrinks towards its mean. Cooling plays the role of λ .

2. **The grazing parameter is quadratic.** For the circle the caustic radius is $R \cos(t/2) = R - \frac{R}{8}t^2 + \dots$, and $\lambda = t^2$ makes the inward motion first order. The network’s internal temperature is $T_l = 1 - \cos \theta_l \approx \theta_l^2/2$ (stage 12): the same reparametrization at the same end. In both, the grazing (deep) regime is where the advance angle is small, and the right coordinate is its square. With T as the parameter, depth is linear in $1/\sqrt{T}$.

3. **Rigidity has a probabilistic twin.**

Proposition 7.1 (proved; elementary). *If every exponential tilt of μ has the same covariance, μ is Gaussian. Indeed $\nabla^2 \Lambda_\mu$ is then constant, so Λ_μ is quadratic. Equivalently, if the Gaussian closure is exact along a full open family of tilts (a “foliation”, not a discrete set), then all cumulants of order ≥ 3 vanish.*

The analogy is in the hypothesis. The paper needs a *full* foliation, because a Cantor family of caustics (Lazutkin) exists for non-ellipses. A closure that is exact on isolated states says nothing; exactness on a continuum forces Gaussianity. The first obstruction is κ_3 , the first odd harmonic of the zonoid.

4. **The method: family invariants pin the high modes.** Sliding periodic families have constant action, which gives one exact equation $C_q(u) = 0$ per high frequency, with derivative equal to the identity plus $O(1/q)$ on the high modes. A Newton scheme on shrinking strips then determines the high-frequency data from the low. Our exact invariants are few:

- the dilation charge (one direction per layer);

- the Euler–Stein ladder, $(1 - r)f_r = (r + 1)(r + 2) \operatorname{tr} f_{r+2}$ (one trace per chaos order);
- the localization variance decomposition $\mathbb{E}A_t + \mathbb{E}a_t \otimes a_t = \operatorname{Cov}$ (one matrix identity per time, automatically satisfied by the closure).

None is diagonal on the flat (incoherent) sector. This is the flex theorem of note XXXVIII seen from the billiard side: the pair state plus our invariants does not determine the omitted fourth-order classes. What did play the role of the missing equations in v56 was extra data, the source-resolved legs, read through the Schur hub (Section 4).

Verdict. The billiard paper supplies a dictionary (support functions, envelopes, positivity, the quadratic grazing parameter), a rigidity principle (only full families constrain), and a method. It supplies no algorithm by itself. The parabolic depth map has no resonances: in Fatou time it is a translation, so the small divisors that drive the billiard proof do not occur in the chain’s depth recursion.

7.2 Packing and exponentially small scales

- **Why sampling cannot see cells.** The mean is a sum over exponentially many cells, each of Gaussian mass about 2^{-n} at the first layer. At floscope’s budget a Monte Carlo pass affords about 6×10^4 samples. That gives per-neuron MSE of order 10^{-6} – 10^{-5} , against the chain’s 1.5×10^{-8} . The “guide the process to the rare event” lesson therefore transfers only in its deterministic form: guided, sticky evolution of the closure’s state.
- **Codegrees.** The Campos–Jenssen–Michelen–Sahasrabudhe insight, that controlling codegrees makes a random-like heuristic exact, has the reading of Section 6: the closure is accurate when the gate gas’s pair couplings are small, and the one coherent coupling is the dilation.
- **Anti-concentration.** The Littlewood–Offord theme, small-ball probabilities of $\sum_a w_a h_a$, is the wall density of stage 12. Its law $(\int Q) \sin(\theta/2)/\sqrt{\pi}$ is an anti-concentration statement for the network’s own fields.

8 Heat flow, slicing, and a memory budget

8.1 A memory budget: negentropy as an integrated localization deficit

Proposition 8.1 (Klartag–Lehec, Corollary 3.2 with Lemma 3.3; restated). *For an isotropic law μ on $\mathbb{R}^n$, with A_t its covariance process under Eldan’s (standard) localization,*

$$D(\mu \parallel \gamma) = \frac{1}{2} \int_0^\infty \left(\frac{n}{1+t} - \mathbb{E} \operatorname{Tr} A_t \right) dt.$$

For the Gaussian $A_t = \operatorname{Id}/(1+t)$ exactly, so the integrand is the localization deficit: how much faster than a Gaussian the law localizes.

Proof. $D(\mu \parallel \gamma) = \frac{1}{2} \int_0^1 \mathbb{E} |v_r|^2 dr$ with $\mathbb{E} v_r \otimes v_r = (\operatorname{Id} - \mathbb{E} \Gamma_r)/(1-r)$, $\Gamma_r = (1+t)A_t$ and $r = t/(1+t)$. Change variables. $\square$

Checked in one dimension by Monte Carlo over the localization path: logistic law, 0.0144 against 0.0142; two-bump mixture, 0.1720 against 0.1728.

Why this matters here. Stage 12 asked for a modular quantity that controls the closure’s error, and named the negentropy of z_l relative to its quasi-free projection as the candidate. Proposition 8.1 turns that negentropy into a time integral of second moments of *localized* laws. By stage 12, the network’s internal clock is a localization time. The chain could therefore estimate its own memory budget from quantities of the same kind it carries: the conditional (quenched) covariances along the localization, the same objects the Schur hub reads. Whether the budget predicts the per-layer closure error is a cheap, truth-free test.

8.2 Shannon–Stam stability: the quenched content as covariance fluctuation

Eldan–Mikulincer (as used by Klartag–Lehec) bound $\int \mathbb{E}|\Gamma_r - \mathbb{E}\Gamma_r|^2 dr/(1-r)$ by the entropy gain of μ under $X \mapsto (X + X')/\sqrt{2}$. That integral is the variance, across localization paths, of the localized covariance, which is exactly the quenched content that separates the Schur hub from its annealed control (Section 4). Its volatility is the third cumulant: $dA_t = \sum_i H_{i,t} dB_{t,i} - A_t dt$ with $H_{i,t} = \kappa_3^{\mu_t}(\cdot, \cdot, A_t^{-1/2} e_i)$ (Bizeul–Klartag–Lehec, eq. (69)). The closure’s ellipsoid is deterministic. The true ellipsoid diffuses, with κ_3 as its diffusion coefficient. This is Klartag’s stochastically evolving ellipsoid with the lattice replaced by the law’s own cumulants.

8.3 Heat-flow contraction and the chaos spectrum

For the Gaussian input, the posterior mean of the output under the observation $X + \sqrt{s}Z$ is $\mathbb{E}[f(X)|X + \sqrt{s}Z]$, the Husimi symbol at the posterior centre. Its variance is

$$\text{Var } \mathbb{E}[f(X) | X + \sqrt{s}Z] = \sum_{k \geq 1} (1+s)^{-k} \|f_k\|^2,$$

with f_k the k -th Wiener chaos component. This is log-convex in s , as Klartag–Ordentlich prove for every log-concave input. The chaos spectrum that v56 grades is thus the sequence of power-series coefficients, in $u = 1/(1+s)$, of one measurable curve: the posterior-mean variance against noise level. The maximal correlation of the Gaussian channel is $(1+s)^{-1/2}$, attained by the first chaos. For the network output, the ratio $\text{Var } \mathbb{E}[f|Y]/\text{Var } f$ falls below $(1+s)^{-1}$ by exactly the share of the higher chaos. Note XLII measured the first-chaos share of deep pre-activations falling with depth, consistent with v56 needing the second and third chaos. One more link: the tilt averages of Bizeul–Klartag–Lehec are, for the Gaussian, $F_f(z) = \mathbb{E}f(X+z)$. Their Taylor tensors are the chaos kernels, and the Song–Zhang criterion turns geometric decay of those kernels into a Poincaré bound. For the chain, the same decay rate controls the error of truncating the chaos grading.

9 What to build next, and what to test

9.1 Builds, ranked by expected value per unit cost

Each item is stated so that it can be pre-registered in the protocol of note XXXIX: cold screen on networks 0–15, then scored with refitted counterterms and judged held-out.

1. The dilation-consistent closure (Section 5).

- *Build.* Carry ε_l . Estimate it from the chain’s own D3 slope, or from the $O(n^2)$ degree-two arc-cosine sum. Overwrite the dilation components of C, D3, D21 and the κ_4 diagonal by the package of Corollary 5.5.

- *Cost.* $O(n^2)$ per layer.
- *Mechanism.* Theorem 5.4: once the package is consistent, the critical mode no longer moves the means at first order, however large its accumulated error.
- *Prediction.* The dilation-mode share of the late covariance error (note XLIV’s c_1 , $0.33 \rightarrow 0.58$) falls substantially, and raw MSE falls by a fraction of the dilation oracle’s value.
- *Bound.* The oracle (true eigenvalue along the true mean at every layer) should be run first, as note XLIV proposed.

2. The remaining L_4 pairings from the existing solve (Section 4).

- *Build.* The $(2, 2)$ path part $\langle D_i, C^{-1}D_j \rangle$ is one product from the solve v56 already does. The off-diagonal pairings $2\langle \kappa_3(ij\cdot), C^{-1}\kappa_3(ij\cdot) \rangle$ and $3\langle D_i, C^{-1}\kappa_3(ij\cdot) \rangle$ need one hub-type contraction of each source with $y_i = C^{-1}Y_i$.
- *Prediction.* The $(3, 1)$ pairing removes a substantial part of the K31 error (74% rms), and its free amplitude refits near 1.
- *Order.* First run the diagnostic: regress the truth’s K31 on the pairing, on note XLI’s dumps, before any chain run.

3. Sticky truth-free constraints. After each layer, project the chain’s state onto the affine set where the exact identities hold, by the minimal Fisher–Rao change: the dilation package, ladder rung 0 ($\mathbb{E}y = 2\text{tr } f_2$, unit by unit) and rung 1 ($\text{tr } f_3 = 0$). The cost is $O(n^2)$. The rung-0 residual alone is a free measurement of what the legs omit (note XLIV, section 8).

4. The memory budget (Proposition 8.1). Estimate the localization deficit from the carried conditional covariances, and test whether it predicts the per-layer injection. This is a diagnostic, not a build.

9.2 Predictions (falsifiable)

1. Successive across-input log-norm increments are uncorrelated, with $\text{Var } \Delta_l \approx \frac{4}{n}(\frac{3}{2} - \frac{J_2(\theta_{l-1})}{2\pi})$. The gain saturates; it does not mean-revert.
2. The truth’s marginal cumulants satisfy, approximately, kurtosis $\approx 2 \times$ the skewness slope (measured 1.6–1.8). The chain’s own satisfy it less well, and the gap tracks the dilation injection.
3. Imposing the dilation package leaves the counterterms’ dilation-sensitive amplitudes (var, D21, K31 at depth) near their unrenormalized values. The package should absorb what they now compensate.
4. The KLS pairings are law terms: their free amplitudes refit at their derived values (1, 2, 3 for the three slices).

9.3 Open problems

1. **Derive the efficiency** $\eta \approx 0.71$. This is the fraction of the across-input log-norm variance that acts as a common scale on the next layer’s bulk. It is a four-point function of the arc-cosine hierarchy.

2. **Integrate the hierarchy along the network's internal clock.** Stage 12 makes each layer a localization step of size $\Delta\Phi_F = 1$. Integrating $-(4\kappa_4 + L_4)$ over the step with the network's own conditional cumulants should give the exact per-layer κ_4 update. Its leading term would be the Schur hub, and the correction terms would come with their weights.
3. **Intrinsic volumes along depth.** $\mathbb{E}f = (V_1(P_+) - V_1(P_-))/\sqrt{2\pi}$. Does the Gaussian closure correspond to a Steiner-type approximation of the tropical polytopes, and is its error a higher intrinsic volume?
4. **A rigidity statement for the chain.** An analogue of the caustic-collar theorem: if a closure is exact on a full neighbourhood of the network's localized states, is the network linear on the support?

Verification

The checks are run by `scripts/verify_stage13.py`; its output is saved next to this note. The network measurements quoted from stage 12 come from `scripts/verify_stage12.py`. The v56 figures come from notes XXXIX, XLI and XLIV on branch `claude/bold-albattani-xgecxi`.

identity	result
1. Eldan cumulant drift (Prop. 4.1, 4.2)	7-point measure in $\mathbb{R}^3$, Gauss–Hermite over one step, Richardson in dt : κ_3 drift $= -3\kappa_3$ to 1.5×10^{-7} ; κ_4 drift $= -(4\kappa_4 + L_4)$ to 2.8×10^{-7} (L_4 is 20%; without it the error is 21%); $L_4(u^4) = 3\langle\kappa_3(uu\cdot), A^{-1}\kappa_3(uu\cdot)\rangle$ exactly
2. First-order cancellation (Thm 5.4)	two-point and uniform scale laws at three (μ, s) : the error falls by 3.8–4.0 per halving of ε , e.g. 1.10×10^{-4} , 2.77×10^{-5} , 6.97×10^{-6}
3. Dilation size law (Prop. 5.2)	$V_l/4 = 0.0051, 0.0058, 0.0064, 0.0068, 0.0072, 0.0075$ at $l = 6, 8, 10, 12, 14, 16$; input radius 4.9×10^{-4} ; per-layer increments 5.4, 2.9, 1.8×10^{-4} at $l = 6, 10, 14$
4. Tropical mean width (Prop. 6.1, Thm 6.2)	$\mathbb{R}^2$, 8 units: $\max \text{relu}(h_A - h_B) - (h_{\text{conv}(A \cup B)} - h_B) = 6 \times 10^{-16}$; $\mathbb{E}f = 0.16269286$ by quadrature and by perimeters
5. Zonoid curvature (Prop. 6.4)	three-component mixture in $\mathbb{R}^2$: Hessian (0.210493, 0.140329, 0.093552) by finite differences and by the wall integral
6. Negentropy budget (Prop. 8.1)	logistic 0.0144 against 0.0142; two-bump 0.1720 against 0.1728 (Monte Carlo over paths, $t \leq 3000$)

References

- [1] Stage 12 note: *the network as its own two-tier system* (this repository, `notes/stage12`).
- [2] Stage 11 note: *tilings, Bratteli diagrams and singular foliations* (`notes/stage11`).
- [3] Working note XLI, *chaos grading and the Schur hub*, and note XXXIX, *the leaderboard system* (branch `claude/bold-albattani-xgecxi`, `notes/chaos-grading`, `notes/leaderboard-system`).
- [4] Working note XLIV, *the fresh-weight theorem*, sections 7–8 (same branch, `notes/fresh-weight`).
- [5] Working note XV, *every computation of the chain* (same branch, `notes/chain-anatomy`); the base chain is 504aldo's open-source factorized third-cumulant chain, v29 (MIT).

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